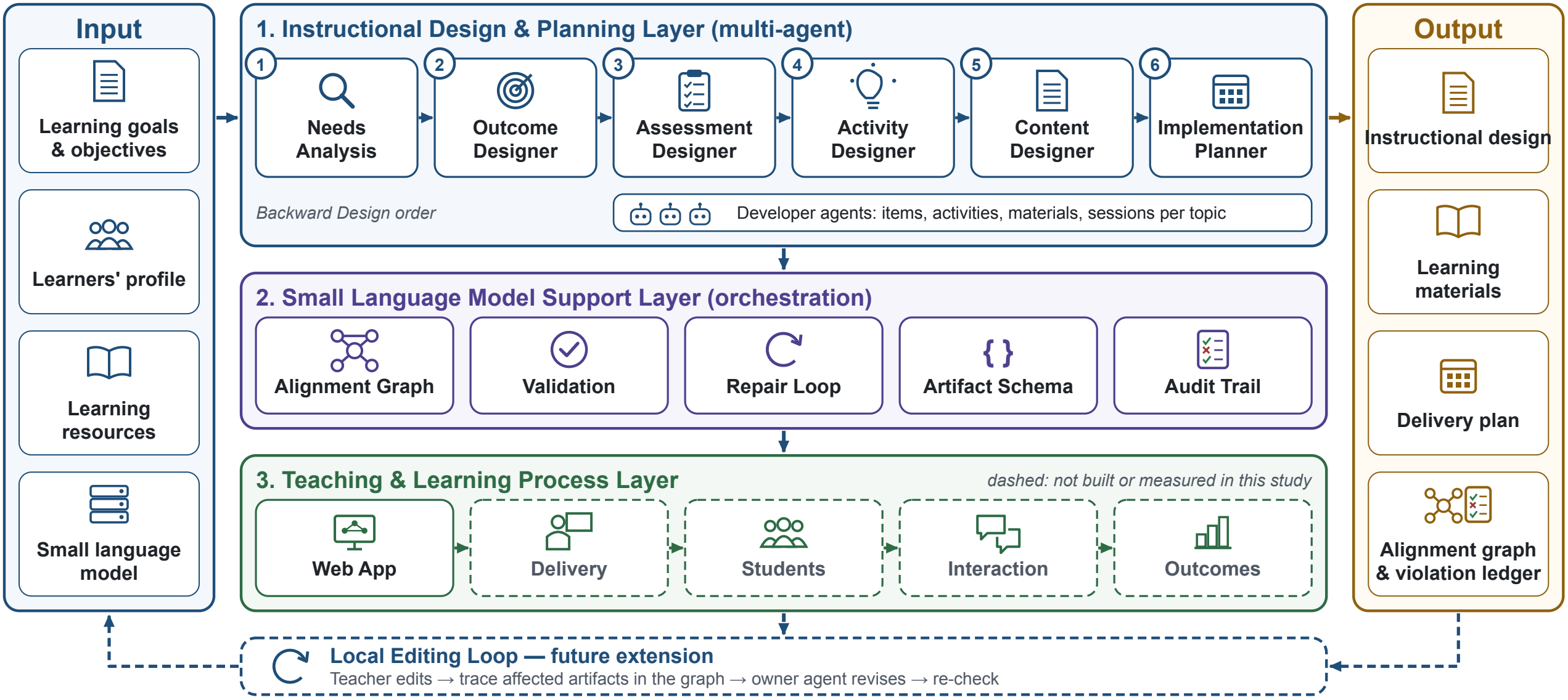

Input
Learning goals & objectives
Learners' profile
Learning resources
Small language model
1. Instructional Design & Planning Layer (multi-agent)
1
Needs Analysis
2
Outcome Designer
3
Assessment Designer
4
Activity Designer
5
Content Designer
6
Implementation Planner
Backward Design order
Developer agents: items, activities, materials, sessions per topic
2. Small Language Model Support Layer (orchestration)
Alignment Graph
Validation
Repair Loop
Artifact Schema
Audit Trail
3. Teaching & Learning Process Layer
dashed: not built or measured in this study
Web App
Delivery
Students
Interaction
Outcomes
Local Editing Loop — future extension
Teacher edits → trace affected artifacts in the graph → owner agent revises → re-check
Output
Instructional design
Learning materials
Delivery plan
Alignment graph & violation ledger